

Electron Desktop UI
TypeScript / React

paper_selection
Query builder
schema editor

QueryBuilder.tsx

paper_selection
Gene relevance
scoring

geneRelevanceScorer.ts

IPC bridge
PROGRESS | LOG | RESULT

python-bridge.ts

output view
Results explorer
CSV / XLSX export

Results.tsx

Python Pipeline
2-4 local workers

paper_selection
NCBI Entrez search
selected PMIDs

query, author, or PMID list

oa_filter
paper_reading
open full text

JATS XML, captions, tables

candidate_discovery
PubTator3 NER
+ HGNC scan

precision floor + deterministic recall

detail_extraction
Gemini schema fields
citations required

custom output columns

review in UI

Validation Framework

validation
HGNC symbol
+ alias check

HGNC snapshot / REST / MyGene.info

validation
source-text
grounding

canonical symbol + aliases

validation
citation cross-ref
SequenceMatcher

citation text exists in paper

validation
variant matching
12+ HGVS patterns

HGVS pattern library

strict gate
confidence ≥ 0.7
 $C = C_g + C_v$

FINAL_VALIDATION_MIN_CONFIDENCE

output_writing
CSV, JSON, XLSX
audit artifacts

deduplicated review files